

CoRoT-1b

R-2625

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_141221 · created 2026-08-28T23:51:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457012.9119 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.015
Transit depth (Rp/Rs)^2	2.05 +/- 0.42 [%]
Orbital Inclination (inc)	87.93 +/- 2.59
Airmass coefficient 1 (a1)	1.038 +/- 0.035
Airmass coefficient 2 (a2)	-0.027 +/- 0.015
Scatter in the residuals of the lightcurve fit is	3.42 %
Best Comparison Star	#1 - [218, 70]
Optimal Method	PSF photometry
Transit Duration (day)	0.1112 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.143 ± 0.015 vs 0.1388 (deviation 0.28σ)
Duration fit vs published	0.1112 d vs 0.1046 d (deviation 1.32σ)
Mid-time O-C	-1.4 min
Depth SNR	9.5
Residual scatter	3.42 %

Light curve

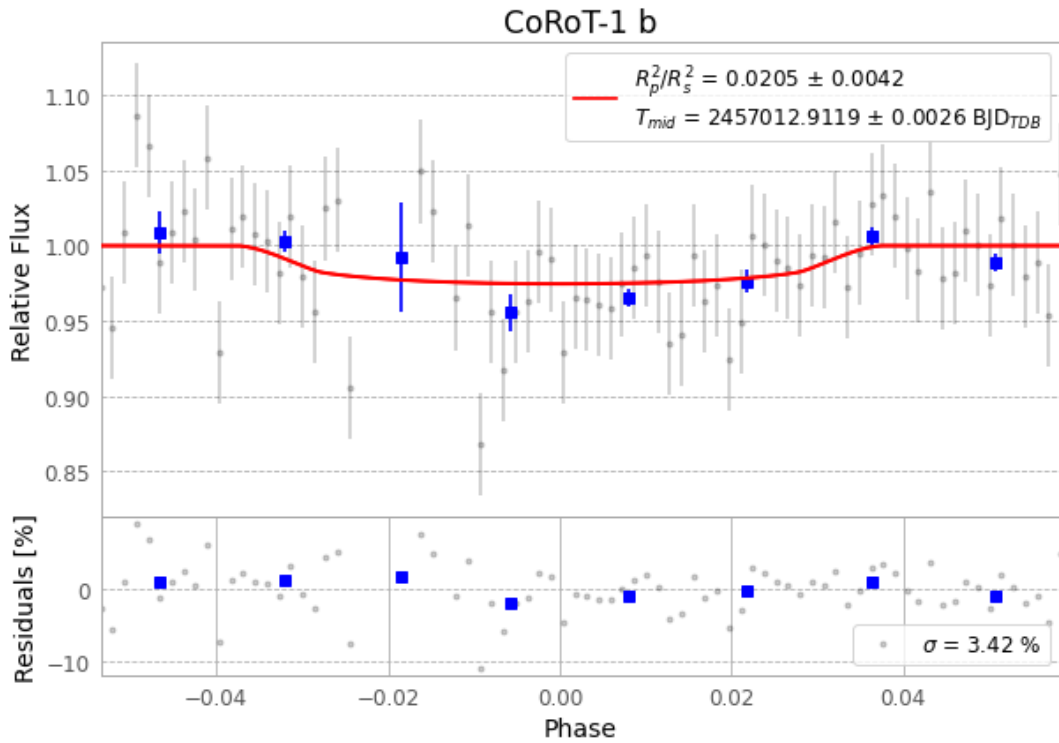


Figure 1 — FinalLightCurve_CoRoT-1 b_2014-12-21.png (SHA-256 42ae2aabf32620b7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [273, 298], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f9a8332b44d5c376...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

82 FITS frames (54 MB), CoRoT-1141221074812.FITS → CoRoT-1141221115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-141221.log (27f40d44cc28...), FinalLightCurve_CoRoT-1 b_2014-12-21.png (42ae2aabf326...), FinalLightCurve_CoRoT-1 b_2014-12-21.csv (8c9a13c1c132...)

Compute resources

Wall time 154.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 23:51Z.